

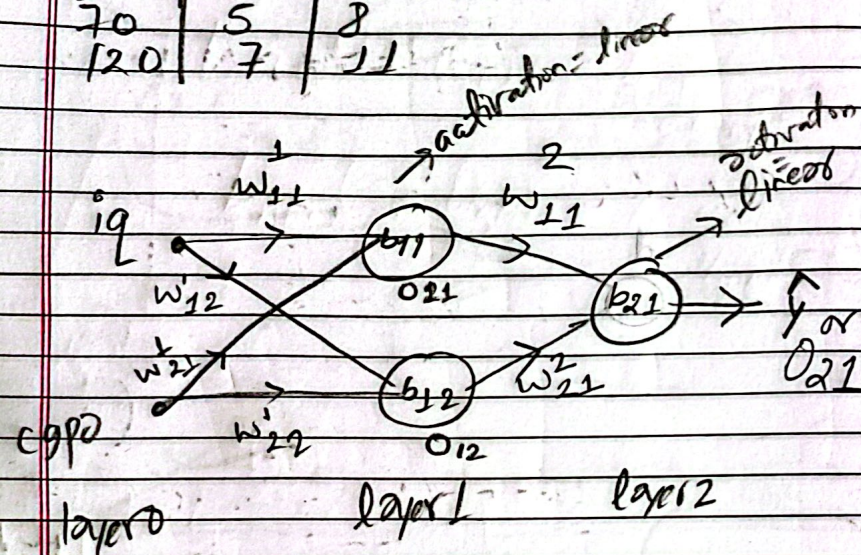
Backpropagation

(Regression problem)

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iq	cgpa	lpa
80	8	3
60	9	5
70	5	8
120	7	11



0) init w, b

(w, b) → random value
or

$w \rightarrow 1, b \rightarrow 0$

Total trainable parameters = $(2 \times 2 + 2) + (2 \times 1 + 1)$
= 9

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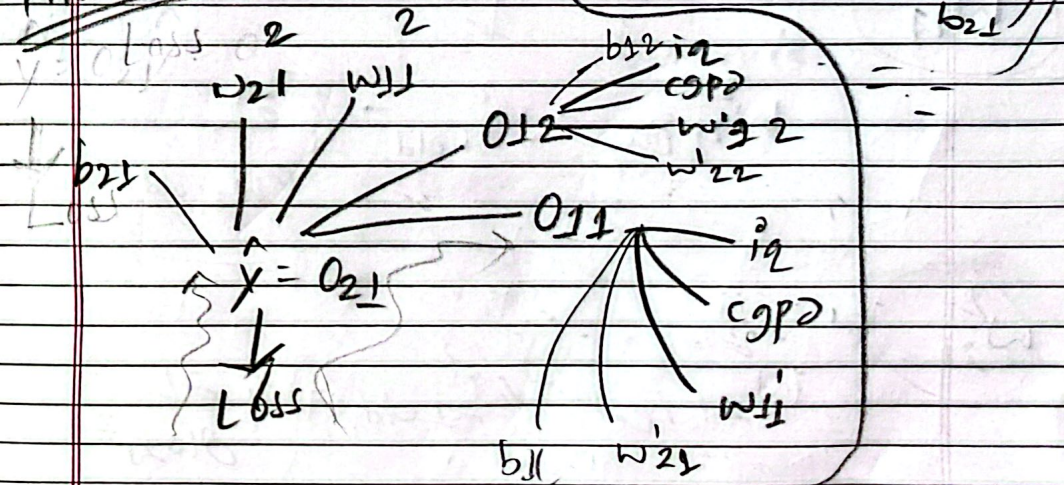
- Steps:
- 0) init w, b
 - 1) In a loop of size of dataset, you select a point (row) → student record
 - 2) predict (lpa) → forward propagation (dot product)
 - 3) choose a loss function & calculate it → LMSE

$$L = (y - \hat{y})^2$$

$$= (y - o_{21})^2$$

$$= (y - (w_{11}^2 o_{11} + b_{21}^2 o_{12} + b_{21}))^2$$

Hierarchy



↓ check previous picture.

update

4) weights & biases (9 trainable parameters)

→ using Gradient Descent for every weight & bias

$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W_{old}}$$

$$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b_{old}}$$

i.e.

W_{11}

known

need to find

$$W_{11}^2_{new} = W_{11}^2_{old} - \eta \frac{\partial L}{\partial W_{11}^2_{old}}$$

known

W_{21}^2

$$W_{21}^2_{new} = W_{21}^2_{old} - \eta \frac{\partial L}{\partial W_{21}^2}$$

& so on.

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Derivative of loss w.r.t weight & biases.

→ total trainable parameters

$$\frac{\partial L}{\partial W_{11}^2}, \frac{\partial L}{\partial W_{21}^2}, \frac{\partial L}{\partial b_{11}} \quad \left| \quad \frac{\partial L}{\partial W_{11}^1}, \frac{\partial L}{\partial W_{21}^1}, \frac{\partial L}{\partial b_{11}} \right|$$

$$\frac{\partial L}{\partial W_{12}^1}, \frac{\partial L}{\partial W_{22}^1}, \frac{\partial L}{\partial b_{12}}$$

Now,

Chain rule

$$1) \rightarrow \frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial W_{11}^2}$$

→ next page.

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} (y - \hat{y})^2$$

$$= -2(y - \hat{y})$$

$$\frac{\partial \hat{y}}{\partial w_{11}^2} = \frac{\partial}{\partial w_{11}^2} [0_{11} w_{11}^2 + 0_{12} w_{21}^2 + b_{21}]$$

$$= 0_{11}$$

$$\text{So, } \frac{\partial L}{\partial w_{11}^2} = (-2(y - \hat{y})) * 0_{11}$$

Similarly,

$$2) \frac{\partial L}{\partial w_{21}^2} = \left(\frac{\partial L}{\partial \hat{y}} \right) * \left(\frac{\partial \hat{y}}{\partial w_{21}^2} \right)$$

$$\frac{\partial L}{\partial \hat{y}} = -2(y - \hat{y})$$

(Previously found)

$$\frac{\partial \hat{y}}{\partial w_{21}^2} = \frac{\partial}{\partial w_{21}^2} [0_{11} w_{11}^2 + 0_{12} w_{21}^2 + b_{21}]$$

$$= 0_{12}$$

$$\text{So, } \frac{\partial L}{\partial w_{21}^2} = (-2(y - \hat{y})) * 0_{12}$$

Similarly,

$$3) \frac{\partial L}{\partial b_{21}} = \left(\frac{\partial L}{\partial \hat{y}} \right) * \left(\frac{\partial \hat{y}}{\partial b_{21}} \right)$$

$$-2(y - \hat{y})$$

$$\frac{\partial \hat{y}}{\partial b_{21}} = \frac{\partial}{\partial b_{21}} [0_{11} w_{11}^2 + 0_{12} w_{21}^2 + b_{21}]$$

$$= 1$$

So,

$$\therefore \frac{\partial L}{\partial b_{21}} = (-2(y - \hat{y})) * 1$$

Now,

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$$4) \frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial w_{11}}$$

$$5) \frac{\partial L}{\partial w'_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial w'_{21}}$$

$$6) \frac{\partial L}{\partial b_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial b_{11}}$$

7)

8) & so on

9)

And finally apply,

$$w_{new} = w_{old} - \eta \times \frac{\partial loss}{\partial w_{old}}$$

$$b_{new} = b_{old} - \eta \times \frac{\partial loss}{\partial b_{old}}$$